Leopold **Mareis**

Statistics · Data Science · Machine Learning

Munich, Germany

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Profile

Statistician finishing a Ph.D. in mathematical statistics at TU Munich (defence December 2026), with a background in dependence modelling for risk and finance, from copula and quantile regression models to efficient estimators with quantified precision. Three years of paid industry projects at Fraunhofer IKS covering model development, validation and client reporting in regulated environments. Works in Python, R, Git and AI-assisted development. Available from January 2027, based in Munich.

Education

Dr. rer. nat. in Mathematical Statistics

Munich, Germany

Technical University of Munich — *Dec. 2021 – Dec. 2026 (expected)*

- Dissertation (submitted): *Asymptotic Inference and Experimenal Design in Linear Causal Models*. Advised by Prof. Dr. Mathias Drton.
- Keywords: Causal Inference, Uncertainty Quantification, Graphical Modelling

M.Sc. in Mathematics in Operations Research (1.3)

Munich, Germany
Delft, Netherlands

Technical University of Munich — *Oct. 2018 – Sep. 2021*

- Thesis: *Vine Copula Based Quantile Regression Including Discrete Components*. Advised by Prof. Dr. Claudia Czado.
- Semester abroad at TU Delft, Netherlands.

B.Sc. in Mathematics (2.0)

Munich, Germany

Technical University of Munich — *Oct. 2015 – Sep. 2018*

- Thesis: *Inexact successive quadratic approximation methods*. Advised by Prof. Dr. Michael Ulbrich.
- Minor: Business and Economics

Experience

Technical University of Munich (MaRDI / NFDI)

Munich, Germany

Scientific Researcher · Research Data & Causal Inference — *Dec. 2024 – Present*

- Conducted statistical research on causal inference and experimental design as lead author, both independently and in collaboration; published and presented at leading venues (CLeaR, PGM, IMS).
- Sole architect and developer of an open research-data platform for graphical modelling and causal inference (NFDI-funded MaRDI initiative), grown to 14k views and 8k downloads.

Fraunhofer Institute for Cognitive Systems (IKS)

Munich, Germany

Scientific Researcher · Reasoned AI Decisions — *Oct. 2021 – Nov. 2024*

- Led the statistical and causal machine-learning analyses in paid industry projects for BIOTRONIK, ADVITOS and Infineon and presented the results to client decision-makers and development boards.
- Built open-source causal inference libraries in Python and R, including the PARCS simulator for benchmarking, documented in an arXiv preprint (arXiv:2309.10514) and reused across projects.
- Advised clients on statistical methodology for developing and certifying AI software in regulated environments. Co-authored a Fraunhofer report on model documentation and validation.

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Munich, Germany

Intern · Corporate Underwriting (Exposure & Loss Modelling) — *Jan. 2019 – Jul. 2019*

- Modelled loss-severity distributions and loss-relief curves in R for exposure analysis.
- Built and automated SAS reporting pipelines (Enterprise Guide, Visual Analytics) on a large property-loss database supporting underwriting.
- Cleaned and consolidated a large claims database to improve data quality for analysis.
- Reviewed facultative reinsurance submissions alongside underwriters.

Agrilution, indoor-farming startup

Munich, Germany

Intern — *Aug. 2018*

Skills

Programming	R, Python, Julia, Git, Docker, LaTeX, Quarto, Vim, command-line workflows, AI-assisted development (Claude Code)
Statistics & ML	Causal inference, semiparametric graphical modelling, GLMs & regression, copulas & quantile regression, experimental design, reinforcement learning, uncertainty quantification
Domains	Financial & insurance risk modelling, clinical trials & biostatistics, industrial R&D
Languages	German (native), English (fluent), French (basic)

Selected Publications

2026 **CLeaR,** Cost-Aware Optimized Front-Door Experimental Design. **L. Mareis**, M. Drton

2026 **PGM,** Semiparametric Inference for Half-Trek Estimators in Linear SEMs. **L. Mareis**, N. Sturma, M. Drton

2025 **Separation & Purification Technology,** Low-Gradient Magnetic Separation in Biotechnology: Parameter Identification and Model-Guided Scale-Up. Meyer et al., incl. **L. Mareis**

2026 **Under review,** Causes and Effectiveness of Gender-Equality Instruments in Organizations. I. Hagerer, **L. Mareis**, D. Schkoda

Full list of publications and talks at lmareis.github.io.

Awards, Teaching & Supervision

2026 **Student Travel Grant,** Conference on Probabilistic Graphical Models (PGM)

Supervision: advised 5 M.Sc. theses and 2 university seminars at TU Munich in graphical modelling, causal inference and machine learning, with applications in biology, finance, insurance and autonomous driving.

Teaching: teaching assistant for 5 courses, including Computational Statistics, Graphical Models in Statistics, and Statistics for Business Administration.

Leadership & Interests

State-certified choir conductor and, since 2024, elected youth representative on the board of the Bavarian Choral Association (BSB), leading rehearsals, organising events and coordinating volunteer teams. Passionate about classical music and choral singing.